



Abhimanyu Susobhanan

Areas of Interest

Gravitational Waves, Pulsars, Data Analysis Methods, Astrophysical Software

Education

Aug 2015–
Sep 2021 **Master of Science in Physics + Doctor of Philosophy in Astrophysics**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*

Thesis Title: [Perspectives in nanohertz gravitational-wave astronomy](#)

Advisor: Prof. Achamveedu Gopakumar

Aug 2008–
May 2012 **Bachelor of Technology in Physical Sciences**, *Department of Earth & Space System Sciences, Indian Institute of Space Science and Technology, Thiruvananthapuram, Kerala, India*

CGPA: 8.27/10

Research & Professional Experience

Nov 2025 –
present **Assistant Professor**, *Indian Institute of Science Education and Research Thiruvananthapuram, Thiruvananthapuram, Kerala, India*

- Noise characterization of the Indian Pulsar Timing Array Data Release 2.
- Preparation of the Indian Pulsar Timing Array Wideband Data Release 2.
- Instructor for *Electrodynamics*
- Mentoring three students and a postdoc
 - Mr Mustafa Bharmal (Master's student, Hyderabad University)
 - Ms Zenia Zuraq (Doctoral student, Indian Institute of Science)
 - Ms Kunjal Vara (Doctoral student, Indian Institute of Science Education and Research Bhopal)
 - Dr Sushmita Agarwal (Postdoctoral Fellow, Tata Institute of Fundamental Research)
- Supervising one student
 - Mr Kiran M R (Doctoral student, Indian Institute of Science Education and Research Thiruvananthapuram)

Apr 2024 –
Oct 2025 **Postdoctoral Fellow**, *Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Hannover, Lower Saxony, Germany*

- Development of the [Vela.jl](#) package for Bayesian pulsar timing & noise analysis.
- Development of advanced data analysis methods for pulsar wideband timing datasets.
- Leading the development team for the [PINT](#) pulsar timing package.
- Mentored one student
 - Ms Réka Desmecht (Master's student, Vrije Universiteit Brussel)

Jun 2022 –
Feb 2024 **Postdoctoral Fellow**, *Center for Gravitation Cosmology and Astrophysics, University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, USA*

- Leading the development team for the [PINT](#) pulsar timing package. Frequentist pulsar timing

- noise characterization using PINT.
 - Efficient methods for computing pulsar timing array signals due to supermassive eccentric binaries, implemented in the [GWecc.jl](#) package.
 - A pilot all-sky single-pulsar Bayesian search for supermassive eccentric binaries using [GWecc.jl](#) in the NANOGrav 12.5-year data of PSR J1909–3744.
 - Multi-messenger targeted search for a supermassive eccentric binary in the radio galaxy 3C 66B in the NANOGrav 12.5-year dataset.
 - The [chimera](#) pipeline to reduce pulsar data obtained using the CHIME telescope, used for integrating CHIME data into NANOGrav.
 - Pulsar timing array signals induced by hyperbolic encounters of supermassive black hole pairs, implemented in the [GW_hyp](#) package.
 - Extension of the wideband timing technique to combine simultaneous multi-band observations.
 - Noise characterization of the InPTA Data Release 1.
 - Substitute instructor for *Survey of Astronomy*.
 - Mentored three students
 - Mr Subhajit Dandapat (Doctoral student, Tata Institute of Fundamental Research)
 - Mr Aman Srivastava (Doctoral student, Indian Institute of Technology Hyderabad)
 - Mr Avinash Kumar Paladi (Master's student, Indian Institute of Space Science and Technology)
- Sep 2021 – **Postdoctoral Fellow**, *National Centre for Radio Astrophysics, Tata Institute of Fundamental Research, Pune, Maharashtra, India*
- Jan 2022
- Led the [InPTA Data Release 1](#) project during its initial stages.
 - Mentored one student
 - Mr Neel Kohle (Master's student, St Xavier's College, Mumbai)
- Aug 2015– **Research Scholar**, *Department of Astronomy & Astrophysics, Tata Institute of Fundamental Research, Mumbai, Maharashtra, India*
- Sep 2021
- The [pinta](#) pipeline to reduce pulsar data obtained using the GMRT telescope.
 - Modeling pulsar timing array signals induced by supermassive eccentric binaries.
 - The *ELL1k* timing model for nearly circular pulsar binaries experiencing significant advance of periastron.
 - A new phasing approach to modeling the optical outbursts of the blazar OJ 287 using its binary black hole central engine model, implemented in the [opha](#) package.
 - An analytic solution to the third post-Newtonian-accurate Kepler equation.
 - Characterization of the 2021 chromatic timing event in PSR J1713+0747.
 - Modeling the jet position angle variations of the blazar OJ 287 using its binary central engine model.
 - Mentored two students
 - Ms Nikita Agarwal (Undergraduate student, Manipal Institute of Technology)
 - Mr Akash Anumalapudi (Undergraduate student, Indian Institute of Technology Bombay)
 - Conducted training sessions on Linux, Python, pulsar data reduction, and pulsar timing at multiple workshops.
 - Teaching assistant for *Astronomy & Astrophysics I*, *Astronomy & Astrophysics II*, and *Electrodynamics II*
 - Public outreach activities including organizing National Science Day and and Frontiers of Science events, and lectures to school and undergraduate students.
- Sep 2012– **Scientist/Engineer**, *Liquid Propulsion Systems Centre, Indian Space Research Organization, Valiyamala, Thiruvananthapuram, Kerala, India*
- Jun 2015
- A searchable internal repository for quality assurance reports named PEARL-DB.
 - Managed a team of junior developers.
 - An internal secure file-sharing service named BHANDAAR.
 - Development and maintenance of the center website, employees' portal, and recruitment portal.

Selected Publications

- 2026* [1] Vidit Singh, Achamveedu Gopakumar, **Abhimanyu Susobhanan**, and Subhajit Dandapat, “*Incorporating post-Newtonian accurate gravitational wave emission effects in pulsar timing array signals from individual supermassive black hole binaries*”, Submitted to Physical Review D
- 2026* [2] Churchill Dwivedi and **Abhimanyu Susobhanan**, “*PSRDISP: A Novel Approach to Modeling Dispersive Processes in Pulsar Timing*”, Submitted to Journal of High Energy Astrophysics
- 2026* [3] Massimo Vaglio, Amodio Carleo, **Abhimanyu Susobhanan**, et al., “*Constraints on Einstein-aether gravity from the precision timing of PSR J1738+0333*”, , Submitted to Physical Review D, arXiv: [2605.01436](https://arxiv.org/abs/2605.01436)
- 2026 [4] Gabriella Agazie, David Kaplan, **Abhimanyu Susobhanan**, et al., “*CHIME-o-Grav: Wideband Timing of Four Millisecond Pulsars from the NANOGrav 15-yr dataset*”, The Astrophysical Journal, 1002, 2, DOI: [10.3847/1538-4357/ae563f](https://doi.org/10.3847/1538-4357/ae563f)
- 2026 [5] **Abhimanyu Susobhanan**, Avinash Kumar Paladi, Réka Desmecht, et al., “*Revisiting wideband pulsar timing measurements*”, Monthly Notices of the Royal Astronomical Society, 547, stag444, DOI: [10.1093/mnras/stag444](https://doi.org/10.1093/mnras/stag444)
- 2026 [6] K Nobleson, ..., **Abhimanyu Susobhanan**, et al., “*The Indian Pulsar Timing Array Data Release 2: II. Customised Single-Pulsar Noise Analysis and Noise Budget*”, Journal of High Energy Astrophysics, 53, 100594, DOI: [10.1016/j.jheap.2026.100594](https://doi.org/10.1016/j.jheap.2026.100594)
- 2025 [7] **Abhimanyu Susobhanan**, “*Bayesian pulsar timing and noise analysis with Vela.jl: the wideband paradigm*”, Physical Review D, 112, 063023, DOI: [10.1103/n3ck-lfdy](https://doi.org/10.1103/n3ck-lfdy)
- 2025 [8] **Abhimanyu Susobhanan** and Rutger van Haasteren, “*Gaussian process representation of dispersion measure noise in pulsar wideband data sets*”, Monthly Notices of the Royal Astronomical Society, 542, 2892–2900, DOI: [10.1093/mnras/staf1422](https://doi.org/10.1093/mnras/staf1422)
- 2025 [9] **Abhimanyu Susobhanan**, “*Bayesian pulsar timing and noise analysis with Vela.jl: an overview*”, The Astrophysical Journal, 980, 165, DOI: [10.3847/1538-4357/adaaec](https://doi.org/10.3847/1538-4357/adaaec)
- 2024 [10] **Abhimanyu Susobhanan**, David Kaplan, Anne Archibald, et al., “*PINT: Maximum-likelihood estimation of pulsar timing noise parameters*”, The Astrophysical Journal, 971, 150, DOI: [10.3847/1538-4357/ad59f7](https://doi.org/10.3847/1538-4357/ad59f7)
- 2024 [11] Subhajit Dandapat, **Abhimanyu Susobhanan**, et al., “*Efficient prescription to search for linear gravitational wave memory from hyperbolic black hole encounters and its application to the NANOGrav 12.5-year dataset*”, Physical Review D, 109, 103018, DOI: [10.1103/PhysRevD.109.103018](https://doi.org/10.1103/PhysRevD.109.103018)
- 2024 [12] Gabriella Agazie, ..., **Abhimanyu Susobhanan**, et al., “*The NANOGrav 12.5-year data set: A computationally efficient eccentric binary search pipeline and constraints on an eccentric supermassive binary candidate in 3C 66B*”, The Astrophysical Journal, 963, 144, DOI: [10.3847/1538-4357/ad1f61](https://doi.org/10.3847/1538-4357/ad1f61)
- 2023 [13] Avinash Kumar Paladi, ..., **Abhimanyu Susobhanan**, et al., “*Multi-band Extension of the Wideband Timing Technique*”, Monthly Notices of the Royal Astronomical Society, 527, 213–231, DOI: [10.1093/mnras/stad3122](https://doi.org/10.1093/mnras/stad3122)
- 2023 [14] Aman Srivastava, ..., **Abhimanyu Susobhanan**, et al., “*Noise analysis in the Indian Pulsar Timing Array Data Release I*”, Physical Review D, 108, 023008, DOI: [10.1103/PhysRevD.108.023008](https://doi.org/10.1103/PhysRevD.108.023008)

- 2023 [15] **Abhimanyu Susobhanan**, “*Post-Newtonian-accurate pulsar timing array signals induced by inspiralling eccentric binaries: accuracy, computational cost, and single-pulsar search*”, *Classical and Quantum Gravity*, 40, 155014, DOI: [10.1088/1361-6382/ace234](https://doi.org/10.1088/1361-6382/ace234)
- 2023 [16] Subhajit Dandapat, Michael Ebersold, **Abhimanyu Susobhanan**, et al., “*Gravitational Waves from Black-Hole Encounters: Prospects for Ground- and Galaxy-Based Observatories*”, *Physical Review D*, 108, 024013, DOI: [10.1103/PhysRevD.108.024013](https://doi.org/10.1103/PhysRevD.108.024013)
- 2022 [17] Pratik Tarafdar, ..., **Abhimanyu Susobhanan**, et al., “*The Indian Pulsar Timing Array: First data release*”, *Publications of the Astronomical Society of Australia*, 39, E053, DOI: [10.1017/pasa.2022.46](https://doi.org/10.1017/pasa.2022.46)
- 2022 [18] K Nobleson, ..., **Abhimanyu Susobhanan**, et al., “*Low-frequency wideband timing of InPTA pulsars observed with the uGMRT*”, *Monthly Notices of the Royal Astronomical Society*, 512, 1234-1243, DOI: [10.1093/mnras/stac532](https://doi.org/10.1093/mnras/stac532)
- 2021 [19] Jaikhomba Singha, ..., **Abhimanyu Susobhanan**, et al., “*Evidence for profile changes in PSR J1713+0747 using the uGMRT*”, *Monthly Notices of the Royal Astronomical Society: Letters*, 507, L57–L61, DOI: [10.1093/mnras/lsab098](https://doi.org/10.1093/mnras/lsab098)
- 2021 [20] Lankeswar Dey, ..., **Abhimanyu Susobhanan**, et al., “*Explaining temporal variations in the jet position angle of blazar OJ 287 using its binary black hole central engine model*”, *Monthly Notices of the Royal Astronomical Society*, 503, 3, 4400–4412, DOI: [10.1093/mnras/stab730](https://doi.org/10.1093/mnras/stab730)
- 2021 [21] **Abhimanyu Susobhanan**, Yogesh Maan, Bhal Chanda Joshi, et al., “*pinta: The uGMRT Data Processing Pipeline for the Indian Pulsar Timing Array*”, *Publications of the Astronomical Society of Australia*, 38, E017, DOI: [10.1017/pasa.2021.12](https://doi.org/10.1017/pasa.2021.12)
- 2020 [22] **Abhimanyu Susobhanan**, Achamveedu Gopakumar, George Hobbs, and Stephen Taylor, “*Pulsar timing array signals induced by black hole binaries in relativistic eccentric orbits*”, *Physical Review D*, 101, 043022, DOI: [10.1103/PhysRevD.101.043022](https://doi.org/10.1103/PhysRevD.101.043022)
- 2018 [23] **Abhimanyu Susobhanan**, Achamveedu Gopakumar, Bhal Chanda Joshi, and Ranjan Kumar, “*Exploring the effect of periastron advance in small-eccentricity binary pulsars*”, *Monthly Notices of the Royal Astronomical Society*, 480, 5260-5271, DOI: [10.1093/mnras/sty2177](https://doi.org/10.1093/mnras/sty2177)
- 2017 [24] Yannick Boetzel, **Abhimanyu Susobhanan**, Achamveedu Gopakumar, Antoine Klein, and Philippe Jetzer, “*Solving post-Newtonian accurate Kepler equation*”, *Physical Review D*, 96, 044011, DOI: [10.1103/PhysRevD.96.044011](https://doi.org/10.1103/PhysRevD.96.044011)

Skills

Programming Languages	C++, Python, C, Julia, Wolfram Language, $\LaTeX$, bash
Astrophysical software	PINT, TEMPO2, ENTERPRISE, PSRCHIVE, DSPSR
Telescope observations	Giant Metre-wave Radio Telescope, Parkes Radio Telescope
Data analysis	Bayesian inference, Data visualization

Languages

Malayalam (native), English, Hindi

Awards, Fellowships, and Grants

- 2026 **ANRF International Travel Support**
- 2019 **Ratanbai Jerajani Award** for the best seminar in the area of Astronomy and Astrophysics at TIFR
- 2019 **Sarojini Damodaran Fellowship** for international travel
- 2006-2012 **National Talent Search Scholarship**

References

Dr. Rutger van Haasteren

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